

A Machine-Checked Finite Reduction for Erdős Problem #287 and a Source-Explicit Large- M Criterion

Clean Public Review + Self-Contained Gate Theory Manual

Version 16.4 – Public Prepublication Candidate R5

Research draft with Lean/Aristotle-assisted formal verification

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Status. Erdős Problem #287 remains open. The unconditional public Lean theorem excludes every exact counterexample with maximum denominator at most 4×10^9 , and the large- M conclusion remains compiled from the explicit `WindowPairSupply` interface. These statements are unchanged by the later analytic research.

Public repository `main` now contains the V15–V19 formal source/algebra banks and imports them through `RequestProject/Main.lean`; the corresponding supplied reports record successful Lean builds of 8118, 8123, 8131, 8135 and 8140 jobs, respectively, with zero errors. The V20 high-quotient shifted-Gram source object is also public, but this paper does not infer a V20 aggregate replay from the existence of that file.

For `Balanced7`, the broad physical obligation

`AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45`

remains open. The V20 high-high-high shifted character Gram is retained as useful formal infrastructure but is *superseded as the controlling frontier*; the earlier schematic HHH inclusion–exclusion closure proof is retracted. The current source-first residual is

`BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45`.

A repaired two-high-projector / double-orthogonality / short- t sieve mechanism survives the present analytic audits, but its physical closure is conditional on the literal Ford–Maynard-(7.23) coefficient identification, prime-box normalization, one shared high-conductor cutoff and the exact external logarithmic prefactor. The factorial/physical comparison remains independently source-open. `Balanced7`, `FCL`, `Gate 2`, `WindowPairSupply`, and Erdős #287 all remain open.

Abstract

Erdős Problem #287 asks whether every representation

$$1 = \frac{1}{n_1} + \cdots + \frac{1}{n_k}, \quad 1 < n_1 < \cdots < n_k,$$

contains a consecutive gap of size at least 3. The strongest unconditional result in this project is machine-checked: there is no exact counterexample with maximum denominator at most 4×10^9 . The remaining large- M conclusion is isolated by a separate machine-checked conditional interface, `WindowPairSupply`. Neither statement is altered by the analytic frontier updates below.

The analytic programme starts from the source-minimal identity

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r, \quad s = \pm 1,$$

and the factorial Euler polarization

$$F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}, \quad a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p).$$

The factorial identity encodes the ordered seven prime boxes exactly, including repeated primes, and the latest synchronized V16 formal bank proves the corresponding algebra and expected-term linearity. The broad physical endpoint **AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45** remains open.

The fixed-degree-seven $1 + 2 + 2 + 2$ analysis reconstructs the pre-Poisson physical geometry. The affine divisor condition

$$q \mid 2mw + s$$

and the second-copy shell recover the apparent $H = X^{11/35}$ loss arithmetically at the natural scale $X^{39/35+o(1)}$. The high-conductor residue source admits an exact character expansion and five-box factorization. The V20 high-high-high shifted Gram is a legitimate formal open socket, but a subsequent hostile audit retracts the earlier schematic HHH inclusion–exclusion closure proof: it did not give one disjoint literal partition of all low-character and overlap cells.

The repaired research architecture works instead with the two original high-character projectors. Exact projector algebra gives an $AA - BA - AB + BB$ decomposition. In the full/full child, double character orthogonality yields simultaneously

$$W - W' = qt, \quad q \mid 2mW' + s, \quad |t| \ll X^{4/35+o(1)}.$$

A source-specific one-dimensional upper sieve for the short t -variable and a Shiu divisor average survive the present analytic audits. A corrected finite bad-character count uses $\#\text{Bad}_q \ll D^2$, leading to

$$C_{\text{var}}(B_0) = \min\{5, \frac{15}{2} - 2B_0, 10 - 4B_0\}.$$

However, physical closure is not yet justified: the seven **Balanced7** weights have not yet been source-identified with one literal Ford–Maynard equation-(7.23) generated packet, their prime-box L^1 normalization is not yet printed, the comparison and analytic decompositions do not yet share a source-pinned cutoff, and the external logarithmic cost C_{ext} is unknown. The exact conditional logarithmic closure criterion is

$$C_{\text{var}}(B_0) > 2C_{\text{ext}}.$$

Accordingly the current first exact residual is the source adapter **BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45**, not an already-closed sieve theorem and not the old HHH Gram. Erdős Problem #287 remains open.

Status at a glance

Item	Status	Meaning
Exact counterexample predicate and $M \leq 4 \times 10^9$ exclusion	PUBLIC LEAN-CHECKED	Strongest unconditional theorem; unchanged by the analytic programme.
WindowPairSupply compiler	PUBLIC LEAN-CHECKED / CONDITIONAL	Eventual effective supply plus the finite theorem implies the public statement.
V15 $\Lambda = \mu * \log$ / affine source / determinant-one algebra	PUBLIC LEAN BANK / SYNCED	Public V15 modules/status are imported; supplied report: 8118 jobs, zero errors.
V16 factorial Euler polarization / local Euler algebra / expected-term linearity	PUBLIC LEAN BANK / SYNCED	Public V16 modules/status are imported; supplied report: 8123 jobs, zero errors.
V17 3221 grouping / source-assisted diagonal / t -range / no-wrap compilers	PUBLIC LEAN BANK / SYNCED	Finite/algebraic/capacity objects only; supplied report: 8131 jobs, zero errors.
V18 normal-form/source-adapter bank	PUBLIC LEAN BANK / SYNCED	Source-blocked normal-form interfaces and firewalls; supplied report: 8135 jobs, zero errors.
V19 pre-Poisson / high-conductor variance / character Gram bank	PUBLIC LEAN BANK / SYNCED	Finite algebra and open interfaces; supplied report: 8140 jobs, zero errors.
V20 HHH shifted-Gram source object	PUBLIC SOURCE / OPEN	Exact open socket is public; no aggregate V20 build is inferred here.
Old HHH inclusion–exclusion closure proof	RETRACTED	The old research proof did not literally partition all low-character/overlap cells.
Two-high-projector reassembly	ALGEBRAIC PASS	Exact $AA - BA - AB + BB$ high-high projector identity.
Double orthogonality	RESEARCH-AUDITED PASS	Gives $W - W' = qt$ and $q \mid 2mW' + s$ in the full/full child.
Short- t one-dimensional sieve	EXTERNALLY AUDITED PASS	Source-specific upper-sieve application; not a new universal theorem.
Shiu divisor average	EXTERNALLY AUDITED PASS	Controls $\sum_{m \sim M} \tau(2W'm + s)$ in the audited progression range.
Bad-character D^2 re-ledger	PASS	Uses $\#\text{Bad}_q \ll D^2$ and the corrected log ledger $C_{\text{var}}(B_0)$.
BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45	SOURCE-OPEN / FIRST RESIDUAL	Must identify the seven physical ω_i with one literal Ford-generated (7.23) packet.
Prime-box L^1 normalization / shared cutoff / external log prefactor	SOURCE-OPEN	Required before the two-high-projector analytic mechanism can close the physical packet.
Two-high-projector five-box sieve	CONDITIONAL	Closes only after the source pins and the strict log inequality $C_{\text{var}}(B_0) > 2C_{\text{ext}}$.
Inverse-sampled high-conductor log variance	CONDITIONAL / NOT CLOSED	Downstream of the same source/normalization conditions.
Factorial/physical comparison	SOURCE-OPEN / DOWNSTREAM	Same-cutoff principal/small-conductor/exceptional-character identification remains separate.
Balanced7 / FCL / Gate 2 / eventual supply / Erdős #287	OPEN	No local packet closure by itself proves the problem.

Status vocabulary. PUBLIC LEAN-CHECKED means a theorem/compiler is present and built in the public formal project. PUBLIC LEAN BANK / SYNCED means the relevant modules and status file are imported by the public aggregate build, with the job count taken from the corresponding supplied report. PUBLIC SOURCE / OPEN means the literal source/interface file is public but the deep estimate is deliberately uninhabited. RESEARCH-AUDITED and EXTERNALLY AUDITED are mathematical audits, not Lean analytic proofs. SOURCE-OPEN means the physical/source identification is missing. CONDITIONAL means the implication is valid only after named antecedents are inhabited. RETRACTED or SUPERSEDED applies to a route/status claim, not automatically to every algebraic identity developed along that route.

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1 The problem and the scope of this note

The public problem asks whether, for distinct integers

$$1 < n_1 < \cdots < n_k, \quad \sum_{i=1}^k \frac{1}{n_i} = 1,$$

one necessarily has

$$\max_{1 \leq i < k} (n_{i+1} - n_i) \geq 3.$$

It is listed as open in the Erdős Problems archive and originates in the sources cited there [1, 2].

For a finite set $A \subset \mathbb{N}$, write

$$N = \min A, \quad M = \max A.$$

A counterexample is a finite set satisfying

$$A \subset \{2, 3, \dots\}, \quad \sum_{a \in A} \frac{1}{a} = 1,$$

and the local gap condition that every nonmaximal $a \in A$ is followed by $a + 1$ or $a + 2$ in A . The Lean development formalizes exactly this predicate and proves the ordered form from it.

This note is deliberately not a chronological research diary. It records only the current controlling finite theorem, the exact conditional closure interface, the source-minimal algebra, the factorial Euler balanced-seven reduction, and the current open analytic obligations. Earlier squarefree/collision routers, Vaughan decompositions, failed provider dictionaries, hostile audits, and superseded residuals belong to the separate technical verification ledger and immutable V15 archive.

2 The machine-checked finite theorem

Theorem 2.1 (Finite exclusion). *There is no exact counterexample to Erdős Problem #287 with*

$$3 \leq M \leq 4,000,000,000.$$

Equivalently, every unit-fraction representation in the problem with maximum denominator at most 4×10^9 has a consecutive gap at least 3.

The theorem is represented in the repository by `no_Erdos287Counterexample_of_max_le_4e9`. Its proof is not an exhaustive enumeration of finite subsets. It is a certificate compiler built from the three ideas below.

2.1 Holes and the adjacent-hole contradiction

For a hypothetical counterexample, call an integer $x \in [N, M]$ a *hole* when $x \notin A$. The `gap ≤ 2` condition implies that two consecutive integers in $[N, M]$ cannot both be holes. This is the formal theorem `Gap2CE.holes_isolated`.

A second elementary fact places the top half of the interval above the minimum:

$$N \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Thus, whenever $M \leq 2x$ and $x + 1 \leq M$, the consecutive pair $x, x + 1$ lies inside the closed window where the adjacent-hole contradiction applies.

2.2 Prime-power windows

For $j \geq 0$, let $C(j)$ denote the largest reduced numerator arising from a nonempty partial reciprocal sum

$$\sum_{s \in S} \frac{1}{s}, \quad \emptyset \neq S \subseteq \{1, \dots, j\}.$$

The formal development uses a certified upper table

$$C_{\text{val}}(0), \dots, C_{\text{val}}(9) = 1, 1, 3, 11, 25, 137, 137, 1019, 2143, 7129.$$

The prime-power window theorem says, schematically, that if p is prime, $e \geq 1$, $p^e \mid x$, and

$$C\left(\left\lfloor \frac{M}{p^e} \right\rfloor\right) < p,$$

then x cannot belong to A . The reason is that the top p -adic layer of the reciprocal sum would force a nonzero numerator to vanish modulo p .

Consequently, if both x and $x + 1$ possess such prime-power divisors, then they are two adjacent holes. This gives the central finite compiler.

Proposition 2.2 (Window-pair blocker). *Suppose there are primes p_u, p_v , positive exponents a_u, a_v , and an integer x such that*

$$\begin{aligned} p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor, \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \\ C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \end{aligned}$$

and

$$M \leq 2x, \quad x + 1 \leq M.$$

Then a gap- ≤ 2 counterexample with maximum M cannot exist.

2.3 The interval-certificate chain

One certificate can exclude an entire interval $L \leq M \leq U$. The divisibility data are fixed at a chosen x , monotonicity controls the windows as M varies, and $U \leq 2x$ keeps the forced holes above the minimum. The repository contains a chain of 34 certificates whose intervals meet end-to-end and cover

$$[3, 4 \times 10^9].$$

All primality side conditions are discharged in the kernel; the construction does not use `native_decide`.

2.4 Sign-sensitive Sophie witnesses and the free maximum blocker

The latest supplied finite bank sharpens the final analytic-to-finite landing and is worth stating because it prevents small endpoint mistakes later. A *plus* Sophie witness at M is a prime q with

$$q \geq 5, \quad M < 3q, \quad 2q + 1 \leq M, \quad 2q + 1 \text{ prime},$$

equivalently $M/3 < q \leq (M - 1)/2$ with the integer endpoints understood literally. A *minus* witness is

$$q \geq 5, \quad M < 3q, \quad 2q \leq M, \quad 2q - 1 \text{ prime},$$

so the correct minus endpoint is

$$q \leq \left\lfloor \frac{M}{2} \right\rfloor.$$

Either witness contradicts an exact counterexample. The hypothesis $q \geq 5$ is not cosmetic: for a prime it is the same as $q > 3$, which is the finite window threshold needed by the cofactor-two obstruction.

There is also a free maximum-divisor blocker. If $q > 3$ is prime, $q \mid M$ and $M < 3q$, then necessarily $M \in \{q, 2q\}$; in either case the top-layer/cofactor-two exclusion contradicts $M \in A$. Thus a counterexample cannot have prime maximum and cannot have $M = 2q$ with $q > 3$ prime. These are finite repairs, not analytic inputs; they should be used when translating positive affine mass into an actual blocker.

3 The exact large- M closure criterion

Definition 3.1 (Window-pair supply). For $M \in \mathbb{N}$, $\text{WindowPairSupply}(M)$ is the assertion that there exist $x, p_u, a_u, p_v, a_v \in \mathbb{N}$ such that

$$\begin{aligned} p_u, p_v \text{ are prime,} \quad a_u, a_v \geq 1, \\ p_u^{a_u} \mid x, \quad p_v^{a_v} \mid x + 1, \\ \left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_u^{a_u}} \right\rfloor\right) < p_u, \\ \left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor \leq 9, \quad C_{\text{val}}\left(\left\lfloor \frac{M}{p_v^{a_v}} \right\rfloor\right) < p_v, \\ M \leq 2x, \quad x + 1 \leq M. \end{aligned}$$

In plain terms, two consecutive integers in the top half of $[1, M]$ each have a prime-power divisor larger than roughly $M/10$, with a small local numerator condition.

Theorem 3.2 (End-to-end conditional compiler). *Assume there is an explicit $M_0 \leq 4 \times 10^9$ such that*

$$\text{WindowPairSupply}(M) \quad \text{for every } M \geq M_0.$$

Then Erdős Problem #287 is true.

This implication is machine-checked as `no_Erdos287Counterexample_of_closure`. The proof splits at 4×10^9 : the finite theorem handles the lower range, while the window-pair blocker handles every larger maximum.

The interface is weaker than a Sophie-Germain prime condition. For example, if a prime q in the interval $M/3 < q \leq (M-1)/2$ has $2q+1$ prime, then one may take $x = 2q$. The analogous minus case uses $x = 2q-1$. The Lean theorem `windowPairSupply_of_sophieWitness` formalizes this implication. The present programme does not claim such Sophie witnesses for every large M .

Part I

Theory and Verification Manual: Gate 0–2

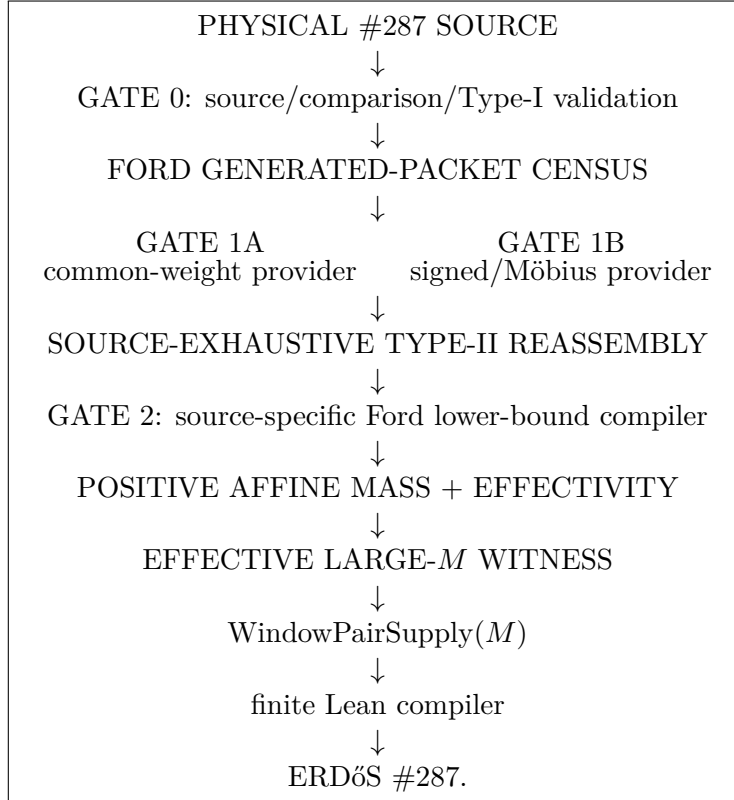
4 What the Gate theory is trying to do

This Part is written for a reader who has never seen the earlier chats, the V15 patch stack, or the repository. The Gate labels are not theorem substitutes: each Gate is a literal mathematical package with an input source, a target theorem, a verification boundary, and specified downstream consequences.

Conceptually, the programme separates four jobs.

- Gate 0.** Construct and validate the physical sequence/comparison class: prime mass, comparison regularity, growth, Type I, and the exact expected-term convention.
- Gate 1.** Supply parity-breaking Type-II estimates for the actual generated packets.
- Gate 1A.** One canonical common-weight moving-source provider.
- Gate 1B.** A complementary signed/Möbius/high-order provider that preserves the arithmetic sign resource through completion/dispersion.
- Gate 2.** Consume a source-exhaustive Type-II package, the Ford arithmetic weight, the exceptional N_2 sector, and one compatible ε_* to run the Ford–Maynard lower-bound endgame.

The intended dependency graph is



(Theory-DAG)

This is a dependency graph, not a proof.

The following non-implications are load-bearing:

Gate 1A closed $\nRightarrow$ Gate 1B closed, Gate 1B closed $\nRightarrow$ Gate 1A closed, Gate 1A + Gate 1B $\nRightarrow$ Full Type II without source exhaustiveness, generated Type II $\nRightarrow$ Gate 2 without Gate-0/ N_2 /source inputs, Balanced7 closed $\nRightarrow$ FCL closed, FCL closed $\nRightarrow$ WindowPairSupply(M), none of the preceding statements alone proves Erdős #287.	(Theory-firewall)
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4.1 Verification vocabulary

Every major object below is assigned one of the following meanings.

Label	Meaning
LEAN-PROVED	Proved by a cited/current Lean kernel replay.
PROVED-FINITE / PROVED-ALGEBRAIC	Exact finite/combinatorial/algebraic mathematics; may or may not also be Lean formalized.
EXTERNALLY-AUDITED	A sourced mathematical theorem or source audit, not formalized here.
CONDITIONAL-COMPILER	The implication is proved, but one or more explicit antecedents remain open.
SOURCE-OPEN	The physical/source identification, normalization or dictionary is missing.
ANALYTIC-OPEN	The source object is sufficiently fixed; a deep estimate remains unproved.
SOURCE-BLOCKED	The literal source object/prefactor needed even to state the final theorem is not yet printed.
CAPACITY-ONLY	Exponent arithmetic passes, but no analytic theorem follows.
FALSE / RETIRED	The literal proposed route is disproved or withdrawn.
SUPERSEDED	Historically valid but no longer the controlling representation.

5 Gate 0: the physical sequence/comparison theorem

5.1 The physical #287 sequence

Fix a nonnegative smooth weight W supported inside the analytic blocker band; the finite blocker itself only needs the corresponding interval to lie in the top half. Define

$$\mathfrak{S}_2 = \prod_{\ell > 2} \frac{\ell(\ell-2)}{(\ell-1)^2}, \quad B(n) = \mathfrak{S}_2 \prod_{\substack{\ell|n \\ \ell > 2}} \frac{\ell-1}{\ell-2},$$

and

$$a_X(n) = W(n/X)(\Lambda(2n-1) + \Lambda(2n+1)),$$

$$b_X(n) = 4W(n/X)B(n), \quad w_X(n) = a_X(n) - b_X(n). \quad (\text{G0-source})$$

The role of b_X is to model the expected local density of the two affine prime forms; w_X is the discrepancy on which Type I/II cancellation is sought.

The elementary comparison algebra includes

$$B(n) = \mathfrak{S}_2 \sum_{\substack{e|n \\ e \text{ odd squarefree}}} \prod_{\ell|e} \frac{1}{\ell-2},$$

and for each fixed d ,

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \frac{d}{\phi(2d)}. \quad (\text{G0-Bmean})$$

Thus one sign has mean $2d/\phi(2d)$ and the two-sign model has mean $4d/\phi(2d)$ at the fixed- d level. Also

$$B(n) \ll \log \log(3n), \quad |w_X(n)| \ll \log X,$$

and, in the present source class,

$$\sum_{n \lesssim X} |w_X(n)| \tau(n) \ll X (\log X)^2. \quad (\text{G0-growth})$$

5.2 Gate-0 target package

Definition 5.1 (Gate-0 package). Gate 0 is *closed for the #287 source at scale X* if the following seven items hold with one common physical normalization.

G0.1 $a_X(n) \geq 0$ and the comparison sequence b_X belongs to the Ford–Maynard ordinary sequence class used in the lower-bound theorem.

G0.2 Prime mass:

$$\sum_p b_X(p) \gg \frac{X}{\log X}.$$

G0.3 Growth/divisor control, including a bound of the form (G0-growth) (or the exact stronger divisor-weight norm required by the chosen Ford invocation).

G0.4 Comparison regularity: the physical b_X satisfies the Ford–Maynard condition (b.2) uniformly over every coagulation region actually used downstream. At the fixed- d level this is compatible with (G0-Bmean); the required uniform publication-level statement must be source-pinned.

G0.5 Literal maximal/weighted Type I:

$$\left| \sum_{m \leq X^{1/2-\varepsilon}} \tau(m)^A \max_I \left| \sum_{\substack{X/2 < mn \leq X \\ n \in I}} w_X(mn) \right| \right| \ll_{A,W,\varepsilon} X (\log X)^{-A}. \quad (\text{G0-I})$$

G0.6 One fixed principal/small-conductor/exceptional-character expected-term convention is used consistently by both the prime source and the comparison source.

G0.7 Even, nonunit, prime-power and boundary strata created by the affine forms $2n \pm 1$ are either included in the source theorem or routed with a proved error.

Current verification. Items G0.1–G0.3 have elementary/source-level proofs in the dossier; the exact finite problem statement and finite blocker are Lean-proved. G0.4 and G0.5 are **SOURCE-OPEN / EXTERNALLY-AUDITED TARGETS** at the exact publication normalization; G0.6 is the same physical comparison convention that later appears in the factorial balanced-seven and Pure5 comparison pins.

What Gate 0 closing buys. It supplies the non-parity hypotheses needed by the Ford lower-bound compiler and removes the possibility that a later Type-II estimate is being applied to the wrong comparison sequence.

What it does not buy. Gate 0 does not prove Type II, parity breaking, Gate 1A, Gate 1B, Gate 2, a large- M witness, or Erdős #287.

6 Gate 1A: canonical common-weight Type-II provider

6.1 Parameters, rows and physical source

At an auxiliary scale X_1 , set

$$\begin{aligned} M &= X_1^{1/3}, & R &= X_1^a, & L &= X_1^b, \\ H &= X_1^{a+2b-2/3}, & K &= \frac{M}{R}, & U &= \frac{L}{H}, & D &= \frac{L^2}{H}. \end{aligned} \quad (\text{G1A-scales})$$

Thus

$$RK = M, \quad DH = L^2, \quad M^2H = RL^2.$$

A row is

$$e = (r, k, m), \quad r \asymp R \text{ prime}, \quad 1 \leq |k| \leq K, \quad m \asymp M, \quad m' = m + kr \asymp M.$$

Choose the canonical affine forms $A_e(t), B_e(t)$ so that

$$\boxed{m'A_e(t) - mB_e(t) = 2k.} \quad (\text{G1A-det})$$

For

$$\rho_\ell(y) = \mathbf{1}_{\ell|y} - \frac{1}{\ell}$$

and bounded prime coefficients b_p, d_q , the *one common physical-weight* source is

$$\boxed{C_e(b, d) = \sum_{t \in \mathbb{Z}} W_D(t) \left(\sum_{p \asymp L} b_p \rho_p(A_e(t)) \right) \left(\sum_{q \asymp L} d_q \rho_q(B_e(t)) \right).} \quad (\text{G1A-source})$$

The word “common” is literal: W_D is one datum. Arbitrary independently chosen weights $W_{D,e}$ are a stronger source class and are not covered by the canonical theorem.

6.2 Gate-1A target theorem

Theorem 6.1 (Canonical common-weight Gate-1A provider – conditional form). *Assume the five source/analytic pins G1A-A through G1A-E below. Then, uniformly for the canonical all- m row family $\mathcal{E}^\sharp$,*

$$\boxed{\sum_{e \in \mathcal{E}^\sharp} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \left(\sum_{p \asymp L} |b_p|^2 \right) \left(\sum_{q \asymp L} |d_q|^2 \right).} \quad (\text{G1A-target})$$

The exact normalized variant used by the formal compiler is equivalent after the banked H^2 normalization bridge.

The theorem is a **CONDITIONAL-COMPILER**: the finite/operator part is substantially banked, but the physical source pins are not all inhabited.

6.3 The five load-bearing pins

G1A-A: Poisson–Bruhat/source transport. After the one-sided physical transform and reciprocity, every retained source state ξ must have moving-prime coefficient

$$c_r(\xi) = B_\xi F_\xi(r/R), \quad (\text{G1A-PB})$$

where F_ξ is a uniformly smooth envelope generated from the one physical W_D , not an arbitrary row weight. The transport must preserve the physical/normalized energy up to a square-summable source error.

STATUS: SOURCE-OPEN / CONDITIONAL PIN.

Without this pin, the later moving-prime participation argument would apply to a convenient model rather than to the physical packet.

G1A-B: absolute source ceiling. The transported physical source must satisfy

$$\mathcal{T}_{\text{abs}} \ll M^2 L^4 X_1^{o(1)}. \quad (\text{G1A-abs})$$

This is the absolute comparator against which the moving-family contraction is measured.

STATUS: SOURCE/ANALYTIC PIN.

G1A-C: fixed-state exclusion and prime participation. A fixed nonprojective state must not absorb the moving prime variable. Smoothness of F_ξ produces a long plateau; the short-interval prime theorem then gives

$$N_\xi \gg R^{3/4-o(1)} \quad (\text{G1A-part})$$

participating prime rows for each retained state. The finite BPP compiler converts this participation into

$$E_{\text{np}} \ll R^{-1/2+o(1)} \mathcal{T}_{\text{abs}}. \quad (\text{G1A-np})$$

The project kernel-checks the finite participation-to-energy implication; it does not formalize the entire physical smooth-envelope/short-interval input.

Status. FINITE COMPILER LEAN-PROVED; the physical participation statement is EXTERNAL / SOURCE-OPEN.

G1A-D: outer-root physical energy and projective routing. The abstract nonprojective energy must be identified with the physical Gate energy, and the projective/axis/diagonal branch must be routed in the same source normalization. The finite ratio-class algebra includes the state diagonal in the zero-curvature branch, and the physical projective mass target is

$$E_{\text{proj}} \ll MHL^4 X_1^{o(1)}. \quad (\text{G1A-proj})$$

Exactly one outer square root converts (G1A-np) to an $R^{-1/4+o(1)}$ contraction. At the three controlling vertices the signed margins are

$$\frac{1}{72}, \quad \frac{1}{24}, \quad \frac{1}{32}. \quad (\text{G1A-margins})$$

Status. FINITE SQUARE-ROOT / PROJECTIVE ALGEBRA BANKED; the physical energy identification is SOURCE-OPEN.

G1A-E: recombination/discarded-sector bound. After all source truncations, exceptional routes and recombinations, the physical error must satisfy

$$E_{\text{rec}} \ll U^{-2+o(1)} M^2 L^4, \quad (\text{G1A-rec})$$

with the source-exact prefactor and without spending the same saving twice. The exponent margins at the controlling vertices are

$$\frac{1}{18}, \quad \frac{1}{18}, \quad \frac{1}{24}.$$

Status. EXPONENT BUDGET BANKED; literal source recombination is SOURCE-OPEN.

6.4 What the BPP mechanism proves and what it does not

The finite BPP mechanism is a local-to-family compiler:

$$\begin{array}{c} \text{smooth moving-prime participation} \\ \Downarrow \\ R^{-1/2+o(1)} \text{ family-energy contraction} \\ \Downarrow \\ R^{-1/4+o(1)} \text{ after one outer root.} \end{array}$$

It does *not* prove the source transport, absolute source ceiling, physical energy identification or recombination estimate. Those are separate statements precisely so that a reader can attack or falsify them independently.

6.5 The source-glue/SB- ν problem

The current source-glue question is whether the same physical packet can simultaneously satisfy the common-weight representation, the BPP state representation and the final routing representation with fibre multiplicity

$$N_\nu = X_1^{o(1)} \quad (\text{G1A-SBnu})$$

in the synchronized two-view model. This is a source-identification frontier rather than a new exponent computation.

Finally,

$$\text{GATE1A_REQUIRED_FOR_287} = \text{UNKNOWN.}$$

The Ford-generated packet census, not programme nomenclature, decides whether a needed #287 packet truly belongs to the canonical common-weight source class.

7 Gate 1B: signed/Möbius parity-breaking provider

7.1 Native centering, sign and complementary-divisor geometry

For coprime d, p ,

$$\rho_{dp}(y) = \rho_d(y)\rho_p(y) + \frac{\rho_d(y)}{p} + \frac{\rho_p(y)}{d}. \quad (\text{G1B-center})$$

If q is squarefree and $p \mid q$ is the distinguished prime, then

$$\mu(q/p) = -\mu(q). \quad (\text{G1B-sign})$$

This is the sign resource that must not be squared away accidentally.

The native complementary-divisor source exposes

$$\boxed{q\ell - uv = 2.} \quad (\text{G1B-det})$$

This determinant-two geometry is separate from the Gate-1A determinant $2k$ and from the #287 $\mu * \log$ determinant-one line.

7.2 The banked one-completion source

On dyadic windows $d \sim D$ and $p \sim P$, the authoritative one-completion coefficient is

$$\boxed{\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p),} \quad (\text{G1B-beta})$$

where μ_D, Λ_P are the physical truncated Möbius and prime weights on the stated windows. The public formal challenge proves the separate linearity/bilinearity of this coefficient and an exact level-of-distribution budget requiring

$$\boxed{\theta \geq \frac{1}{2} + \frac{1}{104}.} \quad (\text{G1B-104})$$

It also proves the implication from an inhabited source-to-convolution-BV interface to the abstract Gate conclusion. Producing that interface from the literal physical source remains open. This is a valid formal provider socket, but it is no longer the first research-level analytic obstruction.

7.3 The Motohashi line

Fix u, ℓ and a base solution to

$$\boxed{uv_0 + 2 = \ell z_0.} \quad (\text{G1B-base})$$

All solutions on the complementary-divisor line are

$$\boxed{v_t = v_0 + \ell t, \quad z_t = z_0 + ut.} \quad (\text{G1B-line})$$

Let H be the physical line length and write $e_H(x) = e^{2\pi i x/H}$. For the exact one-completion coefficient and a smooth cutoff W_1 define

$$\boxed{\mathcal{Z}_{u,\ell}(k) = \sum_t \beta_{D,P}(z_t) W_1(t/H) e_H(-kt).} \quad (\text{G1B-Z})$$

For the five centred defect factors

$$b_5 = \delta_1 * \delta_2 * \delta_3 * \delta_4 * \delta_5$$

and a second physical cutoff W_2 , define

$$\boxed{\mathcal{B}_{u,\ell}(k) = \sum_t b_5(v_t) W_2(t/H) e_H(-kt).} \quad (\text{G1B-B})$$

The Motohashi pairing is

$$\boxed{\mathcal{M}_{u,\ell} = \frac{1}{H} \sum_k \mathcal{Z}_{u,\ell}(k) \overline{\mathcal{B}_{u,\ell}(k)}.} \quad (\text{G1B-M})$$

The current supplied research ledger reports the exact A/B/C source pin, the twisted-defect A/B/C reduction and the fivefold Motohashi iteration as passed source algebra; the fixed-polylogarithmic- k interior is closed modulo the comparison convention.

7.4 Endpoint / character / shift geometry

The earlier endpoint diagonal/off-diagonal identities remain valid source algebra, but they are no longer the first controlling Gate-1B frontier. Later research audits add the following source-preserving reductions.

The determinant-character transfer and the source-minimal nonprincipal-character cross-pairing are recorded as research passes. A positive character-square formulation is a stronger Cauchy sufficient condition, not the source-minimal theorem. High- k Fourier bands can be re-read as weighted short physical shift families. On the shifted source one encounters

$$dp\ell - uv = 2 + ulh, \quad uv \equiv -2 \pmod{\ell},$$

so the character congruence is independent of the shift h even though the determinant is not.

7.5 Current Gate-1B repair frontier

The former

RANKONE-ENDPOINT-U-OFFDIAG45

is retained only as a *historical/superseded endpoint reduction*. A later broad beta-weighted fivefold operator is sufficient but overstrong: generic asymmetric norms do not close it, divisor sparsity does not close it, a free Hilbert/Motohashi operator lift fails, and one beta-side TT^* returns exactly to the earlier rank-two beta frontier.

The current repair candidate is instead

SHIFTED-MAM-TOPBAND45
RESEARCH REPAIR CANDIDATE
PENDING SOURCE-EXACT PROVIDER VALIDATION.

Its motivation is source-theoretic: subtract the already-accounted low-frequency/AP average rather than demand a theorem whose arbitrary shift family contains the native $h = 0$ source as an uncontrolled member.

At the candidate physical scale one has

$$Q = Y^{13/2}, \quad U = Y^4, \quad \theta = \frac{8}{13},$$

and

$$\frac{5}{8} - \frac{8}{13} = \frac{1}{104}.$$

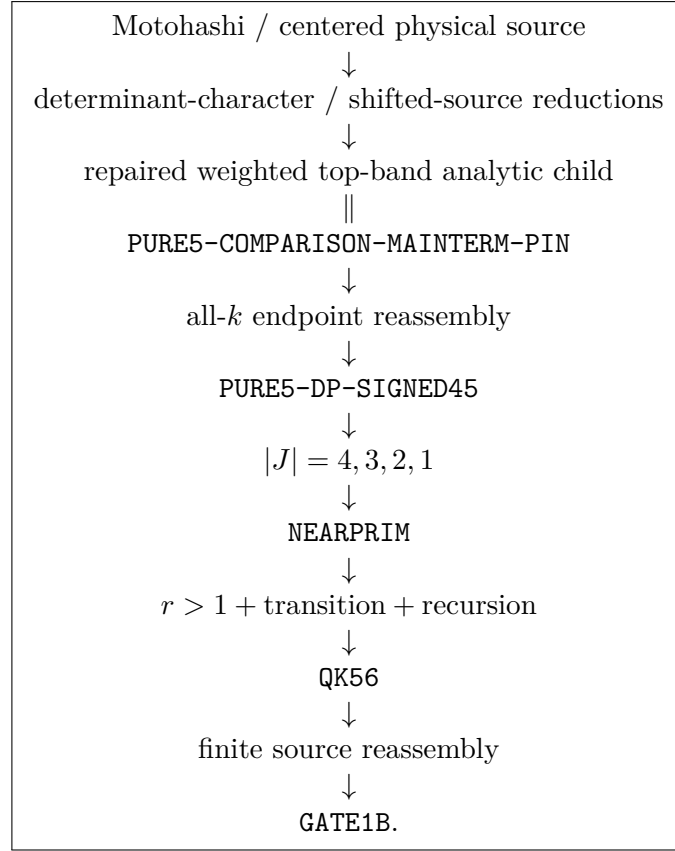
The numerical headroom $8/13$ and $1/104$ is *capacity signal only*; it is not a proved source/provider dictionary. A new gcd child

$$(dp, \ell) > 1$$

must also be routed explicitly.

7.6 Downstream Gate-1B firewall

The present dependency firewall is



The repaired top-band theorem is not claimed proved, and the physical comparison remains source-open.

8 Ford–Maynard generated packets and proof-specific Type II

8.1 Published universal wrapper

The Ford–Maynard lower-bound framework uses a universal Type-II hypothesis. In the present 1/6 specialization a safe sufficient form is

$$\left| \sum_{\substack{(X/2)^\varepsilon < m \leq X^{1/6-\varepsilon} \\ X/2 < mn \leq X}} \xi_m \kappa_n w_X(mn) \right| \ll_A X (\log X)^{-A} \quad (\text{FM-universal-II})$$

for every allowed divisor-bounded coefficient pair. The project does *not* claim that Ford–Maynard publish the weaker proof-specific theorem described next.

8.2 What their proof actually generates

The Ford arithmetic source has

$$G(m; n) = \mathbf{1}_{m \leq n^\gamma} \lambda(m_1) g(\mathbf{v}(m_2; n)), \quad \boxed{H(n) = \sum_{d|n} G(d; n),} \quad (\text{FM-GH})$$

with the canonical rough/smooth split $m = m_1 m_2$. After finite decomposition and condition removal, Proposition 7.22 reaches a fixed-depth product sum (equation (7.23) in the source)

$$\sum_{n=n_1 \cdots n_N \asymp x} w_n x_1(n_1) \cdots x_N(n_N), \quad (\text{FM-7.23})$$

and chooses a product $\prod_{e \in E} n_e$ in the Type-II range. Grouping the selected factors and their complement gives

$$\boxed{\sum_{n=n' n'' \asymp x} Y_1(n') Y_2(n'') w_{n' n''}} \quad (\text{FM-generated-II})$$

The published proof then invokes the universal Type-II hypothesis.

Hence an invocation-by-invocation replacement theorem would be enough for this proof:

every actual generated (FM-generated-II) packet has a proved source-exact provider $\Downarrow$ every Type-II invocation in Proposition 7.22 is covered	(FM-replacement-logic)
--	------------------------

This implication additionally requires a source-exhaustive census and a replacement certificate; it is not obtained merely by identifying some packets.

8.3 The three source obligations

Definition 8.1 (SOURCE-ADAPTER45). For every generated packet, print the literal operator, physical ranges, coefficient class, weight dependence, centering/expected term, gcd/collision restrictions, original prefactor and normalized target, and map it to a proved provider without changing the source class.

Definition 8.2 (FM-TO-GATE-COORDINATE-CENSUS45). Enumerate every Type-II packet actually generated by the Ford decomposition, with enough coordinate information to decide whether it belongs to Gate 1A, Gate 1B or a direct exceptional/standard class.

Definition 8.3 (C_{FM} replacement certificate). Verify invocation by invocation that the generated census covers every use of the universal Type-II hypothesis in the chosen Ford proof, with no omitted packet, no multiplicity loss and no double-spending of an analytic saving.

All three remain open source/reassembly obligations.

8.4 Provider-facing generated families

Family	Literal source/scale	Candidate provider	Why it matches / what is missing
Möbius singleton	one selected $m < X^{1/6}$ fac- tor carries Möbius weight; complement remains generated	Gate 1B signed source	Möbius sign is in the cor- rect physical slot only after a literal completion dictionary; generated outer-factor stabil- ity is still required.
Model singleton	selected factor is model/Mellin-like; Möbius carrier may remain in the complement	QK56 or direct affine completion	provider cannot be inferred from the word “singleton”; na- tive packet dictionary is open.

Family	Literal source/scale	Candidate provider	Why it matches / what is missing
Balanced-seven V-cell	seven fixed prime boxes at $Y = X^{1/7}$	factorial- Ω_7 source, then the fixed-degree-seven attack	factorial Euler encoding is exact; endpoint/3221 analytic theorem and comparison remain.
Pure5/high-order family	five centred defect factors with distinguished-prime/Möbius source	Gate 1B Motohashi/rank-one chain	current first analytic object is the endpoint off-diagonal; high- k and downstream defect/conductor families remain.
Balanced R_9	nine-coordinate Ford-generated leakage cell	fixed-certificate/FCL lane	finite binomial 70 is proved; literal physical $G(d; n)$ specialization is source-open.
Common-weight packets	actual generated source must preserve one physical W_D after scale conversion	Gate 1A	requires exact scale/source dictionary and all five Gate-1A pins.
Remaining fixed- g_* packets	not yet exhaustively censused	TBD from literal source	the finite census/provider assignment itself is still open.

9 Balanced R_9 , fixed-certificate leakage and FCL

The balanced order-nine finite identity is

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70. \quad (\text{R9-70})$$

This is PROVED-FINITE; by itself it is not the physical Ford statement $H(n) = 70$.

To obtain $H(n) = 70$ one must prove that the literal Ford specialization of $G(d; n)$ on the balanced R_9 cell selects exactly the corresponding low/high divisor orders and has the required value of $g(\emptyset)$. That is

$$\boxed{\text{literal } G(d; n) \text{ specialization: SOURCE-OPEN.}} \quad (\text{R9-G-open})$$

Therefore balanced R_9 is a *Ford-generated leakage candidate*, not a completed leakage theorem.

For one fixed positive Ford certificate g_* , define its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Let $\mathcal{P}$ be the desired prime region, $\mathcal{N}_1$ the certificate-sign composite region, $\mathcal{N}_2$ the exceptional collar, and

$$\mathcal{U}_* = (X/2, X] \setminus (\mathcal{P} \cup \mathcal{N}_1 \cup \mathcal{N}_2).$$

The fixed-certificate leakage is

$$\boxed{\mathcal{L}_*(X) = \sum_{n \in \mathcal{U}_*} w_X(n) H_*(n).} \quad (\text{FCL})$$

The N_2 error is

$$\mathcal{E}_{N_2}(X) = \sum_{n \in \mathcal{N}_2} w_X(n) H_*(n).$$

If the total correlation, leakage, N_2 error and comparison error are all small relative to the positive certificate margin $C_*(\varepsilon_*)$, then the fixed-certificate transference gives positive affine prime mass. A sufficient quantitative form is

$$|\mathcal{L}_*(X)| + |\mathcal{E}_{N_2}(X)| \leq \frac{C_*(\varepsilon_*)}{4} B_X, \quad (\text{FCL-margin})$$

together with the corresponding total-correlation/comparison bounds. This is a conditional compiler; it does not itself prove FCL.

Balanced R_9 may contribute to $\mathcal{L}_*$ once the source obligation (R9-G-open) is discharged, but closing that one cell does not close the full leakage family.

10 Gate 2: source-specific Ford–Maynard lower-bound compiler

Fix one compatible

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Gate 2 is the theorem-shaped endgame package:

Theorem 10.1 (Gate-2 lower-bound compiler – conditional). *Assume:*

G2.1 *the Gate-0 source/comparison/Type-I package;*

G2.2 *the literal Ford $G(d;n)/H(n)$ source and its sign properties;*

G2.3 *a source-specific aggregate N_2 estimate;*

G2.4 *one common fixed $\varepsilon_* > 0$ compatible with every source range;*

G2.5 *either the universal Type-II theorem (FM-universal-II), or a complete generated-packet provider package together with **SOURCE-ADAPTER45**, **FM-TO-GATE-COORDINATE-CENSUS45** and C_{FM} ;*

G2.6 *the ordinary/bounded-class firewall and every comparison/exceptional-character normalization needed by the chosen Ford theorem;*

G2.7 *effective versions of all inputs at a computable threshold.*

Then the Ford–Maynard lower-bound geometry yields positive affine mass of the form

$$\boxed{\sum_{p \in \mathcal{P}} a_X(p) \geq (C_*(\varepsilon_*) + o(1)) B_X \gg \frac{X}{\log X}}, \quad (\text{G2-positive})$$

for the physical #287 source and the chosen certificate.

STATUS: CONDITIONAL-COMPILER / NOT ACTIVATED.

The theorem still does not by itself produce **WindowPairSupply**. One additional effective witness extraction theorem must convert the positive affine mass into a Sophie/safe-prime or prime-power blocker configuration, or directly into the **WindowPairSupply**(M) predicate of the finite compiler. That final conversion/effectivity step remains open.

11 Current #287 specialization: $\mu * \log$, factorial Ω_7 , and the 3221 attack

The source-minimal identity used in the current #287 branch is

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r, \quad s = \pm 1. \quad (287\text{-mulog})$$

The latest supplied V15 replay reports the exact three-way split into small- q , small- r , and hard pairs as kernel-checked, together with the determinant-one line

$$qr - 2mn = s.$$

For fixed r, m and one solution,

$$q = q_0 + 2mt, \quad n = n_0 + rt.$$

11.1 Factorial Euler seven-box source

For the balanced-seven cell

$$Q = X^{3/5}, \quad Y = X^{1/7},$$

let

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p), \quad F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}. \quad (\text{Omega7-factorial})$$

Then the degree- $(1, \dots, 1)$ torus coefficient exactly recovers the ordered seven-labelled prime source, including repeated primes:

$$7^7 [z_1 \cdots z_7] F_{\mathbf{z}}(n) = \sum_{\substack{p_1 \cdots p_7 = n \\ p_i \text{ ordered}}} \prod_{i=1}^7 \omega_i(p_i). \quad (\text{Omega7-coeff})$$

The latest supplied V16 factorial replay reports the general/seven-slot polarization, repeated-prime fibre cardinality, formal local Euler algebra and expected-term linearity as Lean-proved in an 8123-job build.

The exact balanced-seven source therefore reduces to

$$\int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, -s\overline{2}) d\mathbf{z}$$

plus the physical comparison identification. The exact broad endpoint theorem is

$$\text{AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45} \quad (\text{Omega7-endpoint})$$

It is strictly narrower than a BV theorem for all class- $\mathcal{C}$ functions, an absolute modulus average, a theorem uniform in every $\mathbf{z}$, or a general smooth-number theorem.

11.2 Why the direct existing Pascadi proof route does not splice

In the audited current v2 proof architecture, a small parameter η is arranged so that

$$Q \leq x^{5/8-100\eta}, \quad y \leq x^\eta.$$

At $Q = x^{3/5}$,

$$\eta \leq \frac{1}{4000},$$

whereas $y = x^{1/7}$ requires

$$\eta \geq \frac{1}{7}.$$

Hence the direct written proof route is nonclosing at the required smoothness scale. This is a parameter/dictionary audit, not a theorem ruling out future extensions.

11.3 The current 1+2+2+2 / 3221 reduction: from pre-Poisson density to the two-high-projector sieve

The fixed-degree-seven grouping is

$$E = X^{1/7}, \quad M = N = L = X^{2/7}, \quad Q = X^{3/5}, \quad EMNL = X.$$

The ordinary three-block comparison sees

$$\frac{4}{7} - \frac{3}{5} = -\frac{1}{35},$$

whereas retaining the actual seventh box in the source-assisted Cauchy step changes the relevant comparison to $Q < ENL$ and yields the capacity margin

$$\frac{5}{7} - \frac{3}{5} = \frac{4}{35}, \quad X^{-2/35+o(1)}$$

at amplitude level. The off-diagonal parameter satisfies

$$e_1 n_1 \ell_1 - e_2 n_2 \ell_2 = qt, \quad |t| \ll X^{4/35+o(1)}, \quad H = X^{11/35}.$$

Physical source reconstruction and pre-Poisson density. The controlling physical relation before completion is

$$\boxed{q \mid 2mw + s.}$$

For fixed (m, w) the admissible moduli are divisors of the nonzero integer $2mw + s$. Together with the second-copy shell of length

$$T = X^{4/35},$$

the natural combinatorial count is

$$M \cdot W_5 \cdot T = X^{39/35+o(1)}.$$

Thus the earlier apparent need to obtain the whole $H = X^{11/35}$ factor from a post-Poisson spectral theorem is superseded: the missing density is already recovered arithmetically at the pre-Poisson level.

The public V19 bank formalizes the affine divisor identity, the second-copy shell, the 39/35 rational exponent ledger, the first-Cauchy consumption of the original $\mu(q)$ sign, the literal high/low character projections, the inverse-sampled high-conductor variance data and the exact character Gram/five-box factorization. These are finite/algebraic interfaces; no deep character-sum estimate is thereby proved.

The V20 HHH Gram and the retraction of the old closure proof. The public V20 file defines the exact surviving high-high-high shifted five-box Gram and leaves its required estimate in an uninhabited analytic structure. That object remains useful formal infrastructure. What is retracted is the earlier *research closure proof* based on a schematic quotient-conductor inclusion–exclusion: it did not supply one literal disjoint partition covering every low- χ , low- $\chi\bar{\xi}$, overlap and moderate-cell intersection. Consequently

3221-HIGHQUOTIENT-FIVEBOX-SHIFTED-GRAM45

is *superseded as the controlling frontier*, not declared false as a formal object.

Two-high-projector repair. Let Bad_q be the low-conductor characters together with the convention-correct exceptional character, and let High_q be the ambient character set minus Bad_q . Then exactly

$$\mathbf{1}_{\text{High}}(\chi)\mathbf{1}_{\text{High}}(\psi) = 1 - \mathbf{1}_{\text{Bad}}(\chi) - \mathbf{1}_{\text{Bad}}(\psi) + \mathbf{1}_{\text{Bad}}(\chi)\mathbf{1}_{\text{Bad}}(\psi),$$

so the physical high-high source decomposes as

$$AA - BA - AB + BB.$$

This avoids quotient-conductor inclusion–exclusion, HHH overlap ambiguity and Burgess-cell double-spending. Status:

TWO-HIGHPROJECTOR-REASSEMBLY45: ALGEBRAIC PASS.

Double orthogonality and the short- t sieve. In the full/full AA child write

$$W = p_1 \cdots p_5, \quad W' = p'_1 \cdots p'_5.$$

Full character orthogonality gives the exact physical relations

$$\boxed{W - W' = qt}, \quad \boxed{q \mid 2mW' + s}, \quad |t| \ll X^{4/35+o(1)}.$$

We record

DOUBLE-ORTHOGONALITY287: PASS.

For fixed admissible (q, W') , the constraint

$$W' + qt = p_1 \cdots p_5$$

forces a rough number in the short t -interval. A standard one-dimensional Selberg upper sieve, with a source-compatible choice such as $z = T^{1/20}$, gives the audited estimate

$$N_t(q, W') \ll \frac{T}{\log X} \frac{q}{\phi(q)},$$

with $z^2 = o(T/\log X)$. This is a source-specific use of standard sieve technology:

ONE-DIMENSIONAL-SIEVE287: EXTERNALLY AUDITED PASS.

For

$$n = 2W'm + s$$

the required divisor average is

$$\sum_{m \sim M} \tau(2W'm + s).$$

The audited Shiu parameter map has ambient scale $\asymp X$, interval length $\asymp X$, modulus $2W' \asymp X^{5/7}$ and residue $s = \pm 1$ coprime to the modulus; the fixed inequality $5/7 < 3/4$ has margin $1/28$. Thus

$$\sum_{m \sim M} \tau(2W'm + s) \ll M \log X$$

up to the harmless local factor $2W'/\phi(2W')$:

SHIU-DIVISOR-AVERAGE287: EXTERNALLY AUDITED PASS.

Corrected bad-character ledger. The safe uniform finite bound is

$$\#\text{Bad}_q \leq \sum_{r \leq D} \phi(r) + O(1) \ll D^2.$$

With

$$D = (\log X)^{B_0},$$

the projected-child logarithmic exponents are

$$AA : -5, \quad BB : -10 + 4B_0, \quad BA/AB : -\frac{15}{2} + 2B_0.$$

Define

$$C_{\text{var}}(B_0) = \min \left\{ 5, \frac{15}{2} - 2B_0, 10 - 4B_0 \right\}.$$

For example, $B_0 = 1$ gives $C_{\text{var}} = 5$, while $B_0 = 2$ gives $C_{\text{var}} = 2$. These are capacity examples only: the physical value of B_0 has not yet been source-pinned.

The current first exact residual is a source adapter. The formal factorial coefficient extraction treats the ω_i abstractly. It does *not* by itself prove the physical statements

$$|\omega_i(p)| \leq 1 \quad \text{or} \quad \sum_{p \sim Y} |\omega_i(p)| \ll \frac{Y}{\log Y}.$$

The first exact residual is therefore

$$\boxed{\text{BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45:} \quad \text{SOURCE-OPEN.}}$$

It must identify the seven physical Balanced7 prime-box weights with the seven literal Ford–Maynard equation-(7.23) generated coefficient slots of one actual packet and print the true coefficient formula rather than impose a guessed one.

Three further source pins must then be discharged:

$$\boxed{\text{BALANCED7-PRIMEBOX-L1-NORMALIZATION45}},$$

requiring the true prime-box L^1 cost (or its actual logarithmic replacement);

$$\boxed{\text{3221-HIGHPROJECTOR-CUTOFF-COMPAT45}},$$

requiring the analytic high-projector and the physical comparison to use the same cutoff $D = (\log X)^{B_0}$; and

$$\boxed{\text{3221-PHYSICAL-LOG-PREFAC45}},$$

which must expose the amplitude-level external logarithmic cost

$$(\log X)^{C_{\text{ext}}}$$

coming from condition removal, Mellin/Perron operations, smooth partitions, dyadic localization and physical source reconstruction.

Exact conditional physical closure criterion. If the outer two-prime Cauchy block satisfies

$$\sum_m |\alpha(m)|^2 \ll M(\log X)^{-2+o(1)},$$

the two-high-projector variance gives $C_{\text{var}}(B_0)$, and the physical source costs $(\log X)^{C_{\text{ext}}}$ at amplitude level, then

$$|S_{3221}^{\text{hi}}| \ll X(\log X)^{-(2+C_{\text{var}}(B_0))/2+C_{\text{ext}}+o(1)}.$$

Hence the exact sufficient condition for $o(X/\log X)$ is

$$C_{\text{var}}(B_0) > 2C_{\text{ext}}.$$

For illustration only: $B_0 = 1$ would require $C_{\text{ext}} < 5/2$, while $B_0 = 2$ would require $C_{\text{ext}} < 1$. These are capacity checks, not current source facts.

The seven boxes must also be source-partitioned explicitly into the outer two-prime Cauchy block and the inner five-prime character/sieve block so that no density factor is charged twice.

Thus the safe current statuses are

3221-TWO-HIGHPROJECTOR-FIVEBOX-SIEVE45
CONDITIONAL ON SOURCE NORMALIZATION / CUTOFF / LOG PREFAC

and

3221-INVERSE-SAMPLED-HIGHCOND-LOGVAR45
CONDITIONAL / NOT CLOSED.

Only after the source adapter, prime-box normalization, shared B_0 , exact C_{ext} , outer/inner ownership and strict log inequality are verified may the comparison identification become the next controlling residual.

The sixth-moment route

3221-HIGHCOND-RESIDUE-SIXTH-MOMENT45

remains a *stronger sufficient research candidate / open*, not the controlling frontier.

12 Verification manual: what is actually proved

12.1 Table 1: machine-checked objects

Object / public bank	Exact checked content	What it does <i>not</i> prove
Finite #287 bank	exact counterexample predicate, hole/window blockers, certificate chain, $M \leq 4 \times 10^9$ exclusion	no large- M theorem
WindowPairSupply compiler	eventual effective supply + finite exclusion $\Rightarrow$ public statement	no inhabitant of eventual supply
V15 (8118 jobs)	$\Lambda = \mu * \log$, affine source, three-way partition, determinant-one line, balanced-seven finite algebra	no prime-distribution estimate
V16 (8123 jobs)	factorial Euler polarization incl. repeated primes, local formal Euler algebra, expected-term linearity, Pascadi parameter ledger	no factorial endpoint or physical comparison match
V17 (8131 jobs)	exact $1 + 2 + 2 + 2$ grouping, source-assisted finite energy, off-diagonal t , no-wrap/capacity compilers	no DI/Kuznetsov estimate; no $X^{o(1)}$ theorem
V18 (8135 jobs)	normal-form/source-adapter interfaces and source firewalls	no physical completed-source inhabitant; no analytic provider
V19 (8140 jobs)	affine divisor identity, first-Cauchy sign consumption, literal low/high character projections, pre-Poisson counts, 39/35 ledger, exact character Gram/five-box factorization	no high-conductor cancellation theorem
V20 public HHH source file	exact surviving HHH Gram object and uninhabited input; four-channel conditional re-assembly compiler	this paper does not infer an aggregate V20 build; the old HHH research closure proof is retracted

12.2 Table 2: externally/research-audited inputs

Input	Status	Scope
Ford–Maynard universal Type-II / lower-bound framework	PUBLISHED INPUT	universal sufficient wrapper; proof-specific replacement still needs invocation-by-invocation source census
Direct Pascadi $(Q, y) = (X^{3/5}, X^{1/7})$ route	AUDITED NONCLOSING	written parameter architecture only; no claim against future extensions
Pre-Poisson divisor-density reduction	RESEARCH / ALGEBRAIC-COMBINATORIAL PASS	recovers the apparent H factor before completion
Two-high-projector algebra	PASS	exact high-high projector reassembly $AA - BA - AB + BB$
Double orthogonality	PASS	yields $W - W' = qt$ and $q \mid 2mW' + s$
One-dimensional short- t sieve	EXTERNALLY AUDITED PASS	source-specific standard upper-sieve application
Shiu divisor average	EXTERNALLY AUDITED PASS	audited progression parameter map
Bad-character D^2 re-ledger	PASS	corrected logarithmic budget $C_{\text{var}}(B_0)$
Old HHH inclusion–exclusion closure proof	RETRACTED	HHH Gram source object remains useful; closure proof did not literally partition all overlap cells
Gate-1B repaired top-band	RESEARCH REPAIR CANDIDATE	pending source-exact provider validation; 8/13 and 1/104 are capacity signal only

12.3 Table 3: open source identifications

Source obligation	Exact missing datum
Gate-0 publication/source pins	literal comparison/Type-I expected-term theorem in the physical normalization
Gate-1A source glue	Poisson–Bruhat/common-weight/outer-root/recombination pins for an actual generated packet
Gate-1B repaired top-band dictionary	source-exact weighted top-band provider plus $(dp, \ell) > 1$ routing
PURE5-COMPARISON-MAINTERM-PIN	physical Pure5 main term / expected-term normalization
Ford generated-(7.23) census	complete list of actual generated coefficient packets and provider assignment
SOURCE-ADAPTER45 and C_{FM}	proof-specific replacement of every Ford Type-II invocation
Balanced R_9 specialization	literal Ford $G(d; n)$ specialization before writing physical $H(n) = 70$
BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45	identify seven physical ω_i with one actual Ford-(7.23) packet
BALANCED7-PRIMEBOX-L1-NORMALIZATION45	print the true prime-box L^1 cost
3221-HIGHPROJECTOR-CUTOFF-COMPAT45	source-pin one cutoff B_0 for analytic and comparison decompositions
3221-PHYSICAL-LOG-PREFAC45	print C_{ext} from every external logarithmic operation
Factorial/physical comparison	same-cutoff principal local density + literal exceptional-character convention
Remaining fixed- g_* census	exhaustive generated leakage/provider assignment

12.4 Table 4: open analytic / conditional theorem slots

Theorem slot	Status	What remains even if supplied
3221-TWO-HIGHPROJECTOR-FIVEBOX-SIEVE45	CONDITIONAL	Requires the source adapter, prime-box normalization, shared B_0 , C_{ext} , outer/inner ownership and $C_{\text{var}}(B_0) > 2C_{\text{ext}}$ before it is a physical closure theorem.
3221-INVERSE-SAMPLED-HIGHCOND-LOGVAR45	CONDITIONAL / OPEN	After closure, physical comparison and Balanced7 reassembly remain.
3221-HIGHCOND-RESIDUE-SIXTH-MOMENT45	STRONGER SUFFICIENT / OPEN	Optional fallback; not the controlling dependency.
Gate-1B repaired weighted top-band	RESEARCH CANDIDATE / OPEN	Still needs source-exact provider test, gcd child, comparison and downstream all- k /Pure5/QK56 reassembly.
Remaining generated packet providers	OPEN	Need complete Ford census and C_{FM} .
FCL / N_2 / Gate-2 analytic inputs	OPEN	Still require effectivity and witness extraction.
Eventual WindowPairSupply	OPEN	Once supplied, the existing finite Lean compiler closes the public statement.

13 Current numbered route to Erdős #287

The current literal route is:

1. prove BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45;
2. print the physical prime-box L^1 normalization, the outer-two/inner-five ownership, one shared cutoff B_0 , and the complete external logarithmic exponent C_{ext} ;
3. verify the strict conditional closure inequality $C_{\text{var}}(B_0) > 2C_{\text{ext}}$ and thereby close the physical two-high-projector packet / inverse-sampled high-conductor log variance;
4. prove the same-cutoff AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45;
5. close the Balanced7–3221 cell;
6. complete the remaining V-cell, M-singleton/model-singleton and generated-packet census;
7. settle the smooth-parity/singleton and remaining fixed- g_* packets;
8. complete SOURCE-ADAPTER45, the Ford coordinate census and C_{FM} ;
9. close fixed-certificate leakage and the N_2 collar;
10. activate Gate 2 at one compatible ε_* ;

11. prove effectivity and extract the required large- M witness;
12. prove eventual `WindowPairSupply`;
13. invoke the existing finite Lean compiler.

Later results may collapse several of these steps. The list is the present literal DAG, not a claim that every intermediate label is mathematically indispensable.

14 How an external reviewer can falsify or improve the programme

A serious review is most useful if it attacks one of the following:

1. a source dictionary that changes the physical packet;
2. a theorem-slot mismatch between an actual generated packet and a claimed provider;
3. a missing normalization or prefactor;
4. a multiplicity or support-enlargement loss;
5. double-spending of a Möbius sign or analytic saving;
6. an incorrect comparison/principal/exceptional-character main term;
7. an incomplete Ford-generated packet census;
8. an invalid invocation-by-invocation replacement of Ford–Maynard’s universal Type II;
9. an endpoint/effectivity gap;
10. a mismatch between a claimed Lean status and the actual kernel/public bank.

Failure of any conditional provider does *not* invalidate the machine-checked finite theorem through 4×10^9 ; it invalidates only the corresponding large- M route.

15 Glossary of recurring programme terms

Gate 0	The physical #287 source/comparison/Type-I package.
Gate 1A	Canonical common-weight moving-source Type-II provider (G1A-source).
Gate 1B	Complementary signed/Möbius/high-order Type-II provider based on (G1B-center)–(G1B-M).
Gate 2	Source-specific Ford–Maynard lower-bound compiler.
BPP	Band-limited prime-participation mechanism converting moving-prime participation to a family-energy saving in Gate 1A.
SB-ν	Synchronized two-view/source-glue fibre condition in the Gate-1A source adapter.
Pure5	Five-centred-defect distinguished-prime source family in the Gate-1B chain.
NearPrim	Near-primitive/full-conductor slice ($r = 1$) of the Gate-1B conductor family.
QK56	Full same-/cross-modulus covariance parent (QK56-parent).
SOURCE-ADAPTER45	Literal source dictionary from actual Ford-generated packets to proved providers.

C_{FM}	Invocation-by-invocation certificate replacing the universal Type-II hypothesis in the chosen Ford proof by generated-packet estimates.
R_9	Balanced nine-coordinate Ford-generated leakage candidate; finite binomial coefficient 70 is proved, physical $G(d;n)$ specialization is not.
FCL	Fixed-certificate leakage/correlation package (FCL).
N_2	Exceptional Ford collar requiring a separate source-specific aggregate bound.
Balanced7	Fixed-degree seven-prime-box V-cell at $Y = X^{1/7}$.
V-cell	A generated packet in the model/factorial seven-prime branch.
Möbius singleton	Generated short Type-II factor carrying the Möbius sign.
fixed-g_*	One fixed positive Ford certificate used in the leakage transference.
WindowPairSupply	The exact large- M adjacent prime-power window predicate consumed by the finite Lean compiler.

16 The source-minimal affine algebra

The current clean narrative starts from the exact identity $\Lambda = \mu * \log$. The supplied V15 Aristotle replay report records a successful 8118-job kernel build of the declarations summarized below. Because public repository **main** has not yet synchronized that replay, their status in this candidate is **LEAN-PROVED IN LATEST V15 REPLAY**, not **PUBLIC LEAN-CHECKED**. No analytic estimate is encoded by these identities.

16.1 The identity $\Lambda = \mu * \log$

For the genuine arithmetic functions,

$$\mu * \log = \mu * (\Lambda * 1) = (\mu * 1) * \Lambda = \Lambda,$$

and hence

$$\Lambda(N) = \sum_{qr=N} \mu(q) \log r.$$

At the affine value $N = 2mn + s$, $s \in \{-1, +1\}$,

$$\Lambda(2mn + s) = \sum_{qr=2mn+s} \mu(q) \log r.$$

The V15 replay also preserves the natural-number sign firewall for the minus case.

16.2 Exact factor partition and determinant-one line

For a cutoff U , factor pairs split disjointly into

$$q \leq U, \quad q > U, r \leq U, \quad q > U, r > U.$$

The hard class is exact rather than heuristic:

$$\sum_{\substack{qr=2mn+s \\ q>U, r>U}} \mu(q) \log r.$$

Every hard pair satisfies

$$qr - 2mn = s, \quad s = \pm 1.$$

This implies

$$(r, 2m) = 1, \quad (q, 2n) = 1.$$

Fixing r, m and one solution (q_0, n_0) , all integer solutions are

$$\boxed{q = q_0 + 2mt, \quad n = n_0 + rt, \quad t \in \mathbb{Z},}$$

with unique t . The V15 replay report records nonvanishing, the coprimality statements, both directions of this parametrization, and multiplicity-one uniqueness as kernel-checked.

The same replay proves only the finite exponent ledger, not an asymptotic estimate. In particular,

$$\frac{1}{6} - \frac{16623}{100000} = \frac{131}{300000} > 0.$$

17 The balanced-seven factorial Euler reduction

Freeze the literal balanced-seven source

$$Q = X^{3/5}, \quad Y = X^{1/7},$$

and

$$\mathcal{B}_{7,s}(X) = \sum_{q \sim Q} \mu(q) \left[\sum_{\substack{p_1, \dots, p_7 \sim Y \\ 2p_1 \cdots p_7 \equiv -s \pmod{q}}} \prod_{i=1}^7 \omega_i(p_i) W(p_1 \cdots p_7 / X) - \mathfrak{M}_{7,s}^{\text{phys}}(q) \right].$$

For odd q , write

$$a_s(q) \equiv -s 2^{-1} \pmod{q}.$$

The target is $o(X/\log X)$. The V15 Lean replay already proves the finite coefficient

$$\sum_{j=0}^3 (-1)^j \binom{7}{j} = -20$$

and the earlier squarefree labelled polarization. The controlling representation below is newer and removes the collision remainder entirely.

17.1 Factorial Euler polarization

For $\mathbf{z} = (z_1, \dots, z_7) \in \mathbb{T}^7$, define

$$a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p)$$

for primes in the actual box, and $a_{\mathbf{z}}(p) = 0$ outside it. Define a multiplicative function $F_{\mathbf{z}}$ by the prime-power data

$$\boxed{F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}, \quad e \geq 0.}$$

This is ordinary multiplicativity; complete multiplicativity is neither required nor desirable here.

Its local Dirichlet factor is

$$\sum_{e \geq 0} F_{\mathbf{z}}(p^e) p^{-es} = \exp(a_{\mathbf{z}}(p) p^{-s}),$$

so the generalized von Mangoldt coefficients satisfy

$$\Lambda_{F_{\mathbf{z}}}(p) = a_{\mathbf{z}}(p) \log p, \quad \Lambda_{F_{\mathbf{z}}}(p^e) = 0 \quad (e \geq 2).$$

Since $|a_{\mathbf{z}}(p)| \leq 1$,

$$|\Lambda_{F_{\mathbf{z}}}(n)| \leq \Lambda(n),$$

so $F_{\mathbf{z}}$ lies in the multiplicative class $\mathcal{C}$ used by Drappeau–Granville–Shao. This is a research/algebraic derivation; the factorial polarization has not yet completed a Lean replay.

17.2 Exact labelled coefficient extraction, including repeated primes

Let

$$n = \prod_p p^{e_p}, \quad \Omega(n) = \sum_p e_p = 7.$$

Then

$$F_{\mathbf{z}}(n) = \prod_p \frac{1}{e_p!} \left(\frac{1}{7} \sum_i z_i \omega_i(p) \right)^{e_p}.$$

The factors $e_p!$ cancel exactly the multiplicity from permuting equal prime occurrences. Therefore

$$7^7 [z_1 \cdots z_7] F_{\mathbf{z}}(n) = \sum_{\substack{p_1 \cdots p_7 = n \\ p_i \text{ ordered}}} \prod_{i=1}^7 \omega_i(p_i).$$

Consequently,

$$\begin{aligned} & \sum_{\substack{p_1, \dots, p_7 \sim Y \\ 2p_1 \cdots p_7 \equiv -s \pmod{q}}} \prod_{i=1}^7 \omega_i(p_i) W(p_1 \cdots p_7 / X) \\ &= 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{\substack{n \asymp X \\ n \equiv a_s(q) \pmod{q}}} F_{\mathbf{z}}(n) W(n/X) d\mathbf{z}. \end{aligned}$$

This identity is exact for repeated primes. Hence the former $O(X^{6/7+o(1)})$ repeated-prime correction is retired from the controlling proof architecture. The earlier collision router remains a valid historical check in the technical dossier.

17.3 Expected-term linearity and the exact comparison pin

Let $\mathcal{E}_q(f; a)$ be any expected-term operator that is linear in f : principal projection, small-conductor subtraction, exceptional-character contribution, or the literal combination adopted by the eventual analytic theorem. Linearity gives

$$7^7 [z_1 \cdots z_7] \mathcal{E}_q(F_{\mathbf{z}}; a) = \mathcal{E}_q(7^7 [z_1 \cdots z_7] F_{\mathbf{z}}; a).$$

Thus polarization itself creates no comparison obstruction.

Define

$$\mathfrak{M}_{7,s}^{\text{fac}}(q) = 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \mathcal{E}_q(F_{\mathbf{z}}; a_s(q)) d\mathbf{z}$$

and

$$\Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, a) = \sum_{\substack{n \asymp X \\ n \equiv a \pmod{q}}} F_{\mathbf{z}}(n) W(n/X) - \mathcal{E}_q(F_{\mathbf{z}}; a).$$

Then

$$\begin{aligned} \mathcal{B}_{7,s}(X) &= 7^7 \int_{\mathbb{T}^7} \overline{z_1 \cdots z_7} \sum_{q \sim X^{3/5}} \mu(q) \Delta_{\mathcal{E}}(F_{\mathbf{z}}; X, q, a_s(q)) d\mathbf{z} \\ &\quad + \sum_{q \sim Q} \mu(q) (\mathfrak{M}_{7,s}^{\text{fac}}(q) - \mathfrak{M}_{7,s}^{\text{phys}}(q)). \end{aligned}$$

There is no repeated-prime remainder.

The comparison problem is therefore not “does polarization commute with the main term?” It does. The remaining source question is the physical identification

$$\mathfrak{M}_{7,s}^{\text{fac}}(q) = \mathfrak{M}_{7,s}^{\text{phys}}(q)$$

at the required aggregate precision: principal local density together with the literal exceptional-character convention. This remains AFFINE287-MULOG-COMPARISON-LOWCOND-MATCH45; it is not closed.

17.4 Broad endpoint obligation and current source-first residual

The broad physical Balanced7 endpoint remains

$$\text{AFFINE287-FACTORIAL-OMEGA7-} \\ \text{SIGNED-ENDPOINT45.}$$

It is not closed.

The current 3221 analysis no longer treats the V20 HHH shifted Gram as the controlling first obstruction. That formal source object remains useful, but the old research inclusion–exclusion closure proof is retracted and the high-quotient HHH cell is superseded as the current frontier. The repaired two-high-projector analytic mechanism is conditional on physical source normalization.

Consequently the present first exact residual is

$$\text{BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45: SOURCE-OPEN.}$$

Only after that adapter prints the physical seven-box coefficients, their true prime-box normalization, outer/inner ownership, one shared cutoff B_0 and the external logarithmic cost C_{ext} can the physical closure test

$$C_{\text{var}}(B_0) > 2C_{\text{ext}}$$

be evaluated. If it passes and the two-high-projector packet closes, the inverse-sampled log variance still has to be reassembled and the independent physical comparison identification proved before Balanced7 is closed.

18 Published distribution results and the exact dictionary audit

Three distinct historical/current facts must not be conflated.

A. Generic smooth-number/factorial route. Drappeau–Granville–Shao work with a broad multiplicative class, while Pascadi’s current arXiv v2 states a smooth-number theorem and remarks on related smooth-supported multiplicative-function technology. The audited written Pascadi parameter architecture uses

$$Q \leq x^{5/8-100\eta}, \quad y \leq x^\eta.$$

At $(Q, y) = (x^{3/5}, x^{1/7})$ this forces simultaneously $\eta \leq 1/4000$ and $\eta \geq 1/7$, so the direct existing route is nonclosing. This is a proof-parameter audit, not a theorem excluding future extensions.

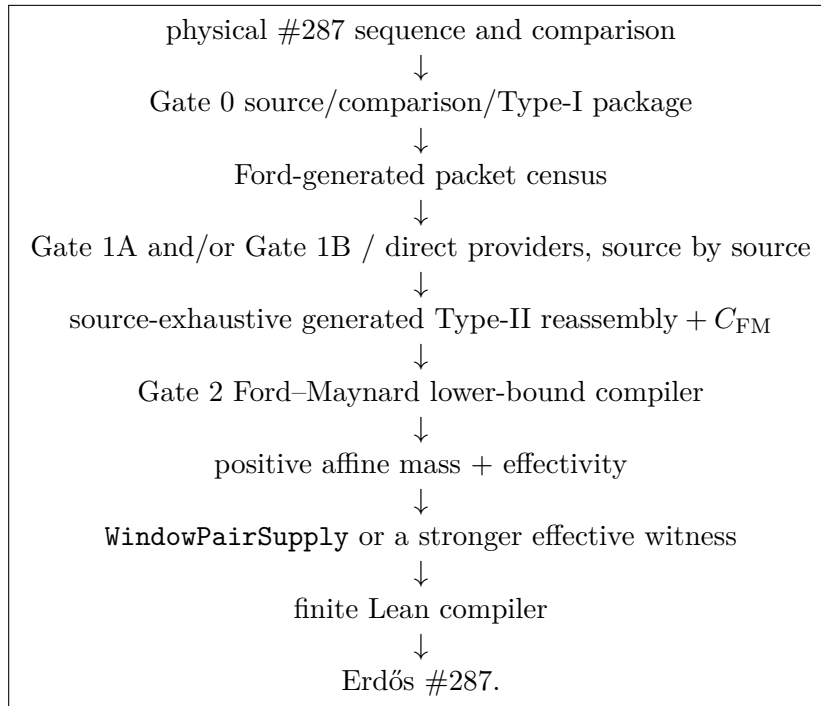
B. DI/Kuznetsov/Kloosterman-fraction provider attempts. Earlier 3221 work explored a literal DI/Kuznetsov source splice and other Kloosterman-fraction black-box lanes. After the physical pre-Poisson source was reconstructed, these became noncontrolling or source-mismatched for the actual packet. Likewise, the demand for a post-Poisson spectral gain of size $H = X^{11/35}$ is superseded: pre-Poisson divisor density already recovers that factor arithmetically.

C. Current two-high-projector sieve mechanism. The present candidate mechanism no longer asks those black boxes to supply the first closure. Its audited analytic ingredients are exact double character orthogonality, a source-specific one-dimensional upper sieve in the short t -variable, a Shiu divisor average, and the corrected D^2 bad-character ledger. The mechanism is not yet a physical closure theorem because the Ford-(7.23) source adapter, prime-box normalization, shared cutoff and external logarithmic prefactor remain source-open.

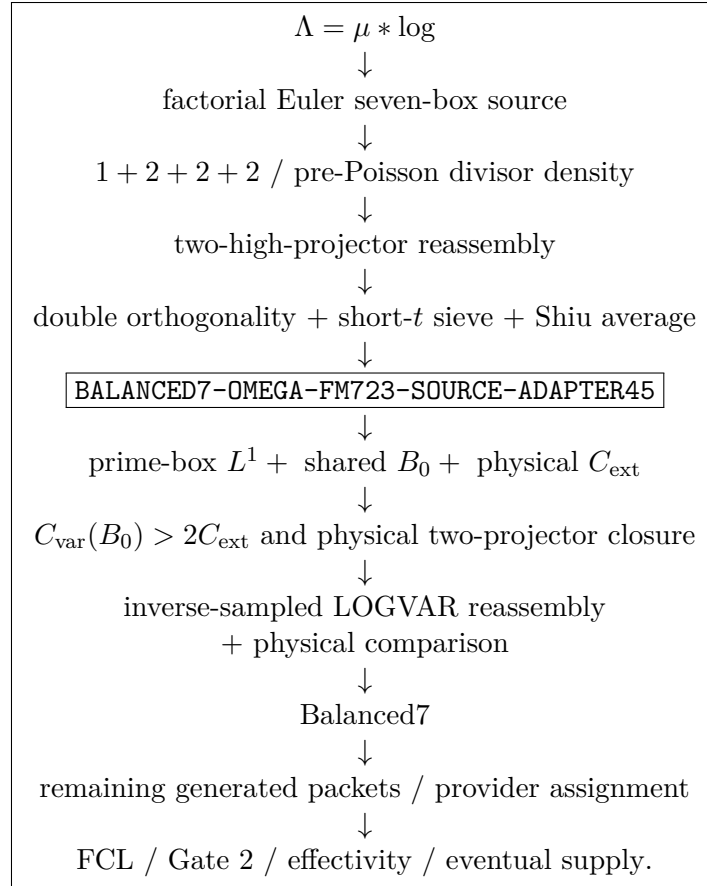
The V20 HHH Gram and sixth-moment infrastructure are retained as formal/provenance tools; they are not the current controlling route.

19 The current end-to-end dependency graph

The Gate spine remains:



The current Balanced7 overlay is



Gate 1B is not assumed globally necessary for every later #287 packet; provider assignment remains census-driven.

The non-implications remain:

Gate 1A closed $\nRightarrow$ Gate 1B closed,

Gate providers $\nRightarrow$ Full generated Type II without source exhaustiveness,

Balanced7 closed $\nRightarrow$ FCL closed, FCL closed $\nRightarrow$ WindowPairSupply.

20 Formal verification and reproducibility

20.1 Public synchronized V15–V19 banks

The public aggregate `RequestProject/Main.lean` currently imports the V15, V16, V17, V18 and V19 source/status modules. The corresponding public reports record:

Layer	Reported build	Scope
V15	8118 jobs, 0 errors	$\mu * \log$, affine source, determinant-one line, finite polarization algebra
V16	8123 jobs, 0 errors	factorial Euler polarization, local Euler algebra, expected-term linearity, parameter ledger
V17	8131 jobs, 0 errors	3221 grouping, finite energy/ t /no-wrap/capacity bank
V18	8135 jobs, 0 errors	normal-form/source-adapter interfaces and source firewalls
V19	8140 jobs, 0 errors	pre-Poisson density, high-conductor variance data, character Gram and compiler interfaces

The reports also record no `sorry` or user axiom in the relevant Lean code and no analytic/source interface being silently inhabited. These are now *public-main synchronized* banks, not merely private replay claims.

20.2 V20 source-object status

The public repository also contains the V20 source object

`RequestProject/Erdos287/HighQuotientShiftedGram3221.lean`.

It defines the exact HHH Gram and an uninhabited analytic input and proves finite anti-vacuity/reassembly statements. At the audit time used for R5, the aggregate file `RequestProject/Main.lean` imports only through V19; no separate V20 successful aggregate-build report was used here. R5 therefore labels V20 as

PUBLIC SOURCE OBJECT / OPEN ANALYTIC,

not as a completed public aggregate replay.

20.3 Research/external analytic status

The two-high-projector identity and the corrected bad-character ledger are mathematical/research passes. Double orthogonality, the short- t upper sieve and the Shiu divisor average are externally/research-audited analytic inputs, not Lean analytic theorems. The old HHH inclusion–exclusion research closure proof is retracted. The current two-high-projector physical closure remains conditional on the source adapter and normalization/cutoff/log-prefactor pins.

20.4 Reproducibility firewall

Every public status is intentionally narrower than the corresponding research ambition: a conditional compiler does not manufacture its antecedent; a finite exponent ledger does not prove an asymptotic estimate; a public source object does not imply its open analytic bound; and a successful supplied build is distinguished from merely planning a future Aristotle run.

21 Conclusion

The unconditional theorem remains finite and machine-checked: there is no exact counterexample to Erdős Problem #287 with maximum denominator at most 4×10^9 . The exact large- M compiler remains `WindowPairSupply`. Neither claim depends on the evolving analytic research frontier.

The broad `Balanced7` endpoint remains

`AFFINE287-FACTORIAL-OMEGA7-SIGNED-ENDPOINT45`,

and it is open. The V20 high-high-high shifted Gram remains useful formal infrastructure, but the old research closure proof based on schematic HHH inclusion–exclusion has been retracted and the HHH Gram is superseded as the controlling frontier.

The current 3221 analysis instead isolates a source-specific sieve mechanism based on exact two-high-projector algebra, double character orthogonality, a short-shift one-dimensional upper sieve, a Shiu divisor average and the corrected D^2 bad-character log ledger. These analytic ingredients survive the present audits. What is not yet source-pinned is the identification of the seven `Balanced7` prime-box coefficients with the literal Ford–Maynard equation-(7.23) generated coefficient slots, together with their exact prime-box normalization, outer-two/inner-five ownership, one shared high-conductor cutoff and the external logarithmic prefactor. Accordingly the current first exact residual is

`BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45`.

The resulting two-projector physical closure remains conditional on

$$C_{\text{var}}(B_0) > 2C_{\text{ext}}.$$

Only after that closure and LOGVAR reassembly does the independent physical comparison identification become the next controlling residual for Balanced7.

Even Balanced7 closure would leave the remaining generated packets, source-exhaustive provider assignment, fixed-certificate leakage, N_2 , Gate 2, effectivity and eventual WindowPairSupply. No local packet theorem alone proves the problem.

Erdős Problem #287 remains open.

Part II

Current Frontier and Provenance Ledger

22 Purpose, coverage baseline and status firewall

Part II restores the *technical depth* of the earlier public dossier without restoring its chronological patch-stack format. Its coverage floor is the V15 public-review snapshot committed as

7fc3797105519c4e6a1669f174ce7394336bdc18],

which published the former 50-page-style Gate-1A/Gate-1B/Ford programme together with the then-current balanced-seven update. That snapshot is used here only as a coverage baseline: when later work changes the controlling route, the later status wins.

Three time/status layers are kept separate throughout:

1. **public/formal bank:** theorem or compiler actually present in the public Lean project;
2. **latest supplied research ledger:** a mathematically audited frontier supplied after the V15 snapshot but not silently promoted to a public Lean or published theorem;
3. **open/source interfaces:** deliberately unproved analytic or source-identification obligations.

This is important for Gate 1B. The public formal socket still records the one-completion coefficient

$$\beta_{D,P}(d,p) = \mu_D(d)\Lambda_P(p)$$

and the exact 1/104 distribution-margin budget. That socket is neither false nor erased. However, the latest supplied analytic work has moved beyond it to a Motohashi/rank-one frontier. The formal socket is therefore a *banked abstraction/alternative provider interface*, while the rank-one chain below is the current research frontier.

Detailed dead-route archaeology remains outside the main narrative. What is restored here is every current component needed to reconstruct the proof programme: Gate 1A source class, Gate 1B current frontier and downstream firewall, the Ford-generated packet census, balanced- R_9 /fixed-certificate leakage, Gate 2, and the relation of the current factorial- Ω_7 source to those providers.

23 Gate 1A: current canonical-provider ledger

Gate 1A is a conditional theorem for one canonical common-weight physical operator. It is not a universal theorem for arbitrary row-dependent source weights and it is not assumed globally necessary for Erdős #287.

For a row

$$e = (r, k, m), \quad m' = m + kr,$$

the affine forms satisfy

$$m'A_e(t) - mB_e(t) = 2k,$$

and the canonical physical operator is

$$C_e(b, d) = \sum_t W_D(t) \left(\sum_{p \preceq L} b_p \rho_p(A_e(t)) \right) \left(\sum_{q \preceq L} d_q \rho_q(B_e(t)) \right).$$

The conditional target remains

$$\sum_{e \in \mathcal{E}^\#} |C_e(b, d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)} \sum_{p \preceq L} |b_p|^2 \sum_{q \preceq L} |d_q|^2,$$

with the same normalization conventions as in Part I.

23.1 What is banked and what is still external

Gate-1A component	Status	Current interpretation
Determinant/row algebra and all- m row family	BANKED	The theorem is not restricted to clean- P_3 rows.
Canonical common physical weight W_D	FINITE/OPERATOR	
Scale ledger, row-energy conservation and normalization bridge	SOURCE FIREWALL	Later row dependence must be deterministically derived from this one datum.
Finite BPP participation-to-energy compiler	BANKED	Includes the H^2 conversion and the signed vertex margins.
Projective ratio-class/axis algebra	BANKED	Participation implies a moving-family energy saving once the physical participation input is supplied.
Outer-square-root algebra and exponent budget	BANKED FINITE	The state diagonal is routed inside the zero-curvature/projective branch rather than discarded.
Poisson–Bruhat transport/source ceiling	ALGEBRA	The formal compiler checks the square-root conversion but does not manufacture the physical energy pin.
Fixed-state exclusion and prime participation	BANKED	Must preserve the literal physical packet and absolute comparator.
Outer-root physical energy identification	EXTERNAL/SOURCE	The finite compiler is present; the physical smooth-envelope/short-interval input is external.
Projective/diagonal physical source routing	PIN	Identifies the abstract BPP energy with the actual Gate energy.
Literal recombination estimate	SOURCE OPEN	Finite algebra exists; the physical source ceiling must be supplied at target scale.
Arbitrary edge-dependent D_2 family	SOURCE OPEN	Controls discarded/excision/recombination errors without spending the same saving twice.
	OUTSIDE CANONICAL	Requires an upstream Gate-0 adapter or finite-template/nuclear reduction.

The current source-glue issue can be summarized as a synchronized two-view/fibre problem: the same physical packet has to satisfy the common-weight representation, the BPP participation representation and the final source-routing representation simultaneously. We refer to this compactly as the *SB- ν /source-glue interface*. It is a source-identification problem, not an exponent deficit in the already-banked finite operator algebra.

23.2 When Gate 1A is required for #287

The correct logical statement is

GATE1A_REQUIRED is decided by the literal Ford-generated packet census.

An actual generated packet may use Gate 1A only after a scale/source dictionary supplies:

- the relation between the Gate ambient scale X_1 and the #287 scale X ;
- the identification of the physical bilinear variables with the Gate row variables;
- the induced R, L, H, K, U, D ranges;
- preservation of the one common physical weight, or a proved finite-template/nuclear reduction with acceptable total cost;

- survival of the binding fixed-power margin after truncation, multiplicity and scale conversion.

No global Gate-1A closure is assumed if the generated census never routes a needed packet to this source class.

24 Gate 1B: current provenance-aware frontier

24.1 Banked/source background

The one-completion source

$$\beta_{D,P}(d, p) = \mu_D(d) \Lambda_P(p)$$

and the exact 1/104 dyadic budget remain valid formal/source architecture. Motohashi / fivefold source decompositions and the centered two-frequency/character reductions remain research-bank inputs.

24.2 What was superseded

The former

RANKONE-ENDPOINT-U-OFFDIAG45

is historical/superseded as the current first frontier. The later broad beta-weighted fivefold operator is a sufficient but overstrong reduction: generic asymmetric norms and divisor sparsity do not close it; a free operator/Hilbert Motohashi lift fails; one beta-side TT^* is an exact representation loop back to the rank-two beta frontier. A uniform $K^{-\delta}$ high- k shortcut is retired.

24.3 Current repair candidate

The current conservative research status is

SHIFTED-MAM-TOPBAND45
RESEARCH REPAIR CANDIDATE / PENDING SOURCE-EXACT PROVIDER TEST.

The repair subtracts the already-accounted low-frequency/AP average instead of asking for a theorem containing the native $h = 0$ source as an arbitrary member of a shift supremum.

At the candidate level

$$Q = Y^{13/2}, \quad U = Y^4, \quad \theta = \frac{8}{13}, \quad \frac{5}{8} - \frac{8}{13} = \frac{1}{104}.$$

This is a capacity signal, not a proved published-provider dictionary. The child $(dp, \ell) > 1$ must be routed explicitly.

The physical comparison remains source-open. The downstream firewall is:

repaired top-band || comparison
↓
all- k reassembly
↓
Pure5 $\rightarrow |J| = 4, 3, 2, 1$
↓
NearPrim
↓
 $r > 1 + \text{transition} + \text{recursion}$
↓
QK56
↓
finite source reassembly
↓
Gate1B.

No downstream node is promoted by the existence of the capacity margin alone.

25 Ford–Maynard generated packets: census before provider

The Ford–Maynard universal Type-II hypothesis is a clean published sufficient wrapper, but Proposition 7.22 does not invoke arbitrary coefficient pairs out of thin air. After a finite decomposition it generates the grouped equation-(7.23) family

$$\sum_{n=n'n''\asymp x} Y_1(n')Y_2(n'')w_{n'n''}.$$

A proof-specific replacement is therefore legitimate only after two source obligations:

1. a *literal generated-packet census* listing every coefficient family produced by the proof;
2. a replacement certificate C_{FM} proving that every Type-II invocation is covered by one proved provider after all Mellin/Perron, support and norm transformations.

The programme labels

SOURCE-ADAPTER45,
FM-TO-GATE-COORDINATE-CENSUS45

refer to this source dictionary. Neither can be replaced by the slogan “the packet looks Type-II”.

25.1 Current provider-facing census

Generated family	Literal status	Candidate provider	Current blocker
Möbius singleton	source-specific generated packet	Gate 1B / signed source	literal dictionary and stability under gener- ated outer factors
Model singleton	source-specific generated packet	QK56/direct affine completion	native packet dictio- nary; short Möbius car- rier may sit in the com- plement
Balanced-seven V-cell	explicit fixed-degree source	factorial- Ω_7 endpoint theorem	signed endpoint + physical comparison identification
Balanced R_9 leakage	Ford-generated leakage candidate	fixed-certificate/FCL lane	literal $G(d; n)$ special- ization and source re- assembly
Common- weight/edge packets	census-dependent	Gate 1A only after exact match	scale/source adapter and common-weight preservation
Pure-five/high-order packets	Gate-1B source family	Motohashi/rank-one chain	endpoint off-diagonal, comparison, high- k , downstream strata
Remaining fixed- g_* packets	not exhaustively censused	TBD from literal packet shape	finite census and provider assignment

This table encodes the main logical correction:

provider assignment follows the generated source; it is not imposed globally.

In particular, Gate 1A is neither deleted nor assumed universally necessary.

26 Balanced R_9 , fixed-certificate leakage and the Gate-2 landing

The balanced order-nine identity

$$\sum_{j=0}^4 (-1)^j \binom{9}{j} = 70$$

is a genuine finite/binomial fact. The formal bank also records the complementary alternating identity. What is *not* automatically proved is that Ford's actual arithmetic weight $G(d; n)$ selects exactly this divisor-order specialization on the physical balanced- R_9 cell.

Therefore the correct status is:

balanced R_9 = Ford-generated leakage candidate,

with

$H(n) = 70$ only after the literal $G(d; n)$ specialization is proved.

The old R_9 death certificate and the abandoned unbalanced-1/5 redesign are not revived.

The fixed-certificate architecture uses one explicit positive Ford certificate g_* and its arithmetic weight

$$H_*(n) = \sum_{d|n} G_*(d; n).$$

Its transference identity separates:

- total correlation;
- leakage outside the prime/main-composite regions;
- the exceptional N_2 collar;
- the positive comparison margin.

Balanced R_9 is one generated leakage family that can feed this architecture once the physical source dictionary is proved; it is not the whole fixed-certificate leakage problem.

26.1 Gate 2 consumes, rather than invents, the packet bounds

Gate 2 is the Ford–Maynard lower-bound/endgame compiler. It fixes one compatible ε_* and the shrunk parameter triple

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*, \varepsilon_*, \frac{1}{6} - 2\varepsilon_* \right).$$

Its source-specific replay requires:

1. Gate-0 comparison regularity, growth and exact Type-I input;
2. the literal $G(d; n)/H(n)$ dictionary and the Ford-generated packet census;
3. the source-specific aggregate N_2 estimate;
4. one compatible fixed ε_* ;
5. either the universal Type-II wrapper, or all generated-(7.23) packet estimates together with SOURCE-ADAPTER45 and C_{FM} .

Thus the correct closure order is

packet census $\rightarrow$ Gate providers $\rightarrow$ generated Type-II reassembly $\rightarrow$ Gate 2 lower bound,

not Gate 2 followed by a later packet census.

27 Current #287 fixed-degree-seven direction inside the gate programme

The controlling factorial source is

$$F_{\mathbf{z}}(p^e) = \frac{a_{\mathbf{z}}(p)^e}{e!}, \qquad a_{\mathbf{z}}(p) = \frac{1}{7} \sum_{i=1}^7 z_i \omega_i(p).$$

The broad factorial endpoint and the physical comparison remain open.

The historical route through a DI/Kuznetsov literal splice and the later V20 HHH shifted Gram are both retained as provenance/infrastructure, but neither is the present first frontier. The current analytic mechanism uses the two original high-character projectors:

$$AA - BA - AB + BB.$$

Double orthogonality in AA gives

$$W - W' = qt, \qquad q \mid 2mW' + s, \qquad |t| \ll X^{4/35+o(1)}.$$

The short- t one-dimensional sieve and the Shiu divisor average are externally audited passes. The corrected low-character count is $\#\text{Bad}_q \ll D^2$, giving

$$C_{\text{var}}(B_0) = \min \left\{ 5, \frac{15}{2} - 2B_0, 10 - 4B_0 \right\}.$$

The present first exact residual is not a new analytic black box but the source adapter

BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45.

It must print the seven literal Ford-(7.23) coefficient slots. The accompanying source pins are the prime-box L^1 normalization, outer-two/inner-five ownership, one shared cutoff B_0 and the external logarithmic prefactor C_{ext} . Only if

$$C_{\text{var}}(B_0) > 2C_{\text{ext}}$$

does the candidate two-high-projector mechanism close the physical high-conductor packet and permit LOGVAR reassembly. Comparison is downstream of that test.

Thus

Balanced7 closed $\not\Rightarrow$ FCL closed $\not\Rightarrow$ WindowPairSupply.

28 Master current-status ledger

Node	Current status	Publication interpretation
Finite exclusion $M \leq 4 \times 10^9$ WindowPairSupply compiler	PUBLIC LEAN-CHECKED PUBLIC LEAN-CHECKED / CONDITIONAL	strongest unconditional theorem exact global large- M interface
V15–V19 formal banks	PUBLIC-MAIN SYNCHRONIZED	reported builds 8118/8123/8131/8135/8140, zero errors
V20 HHH Gram source object	PUBLIC SOURCE / OPEN	exact open object; no aggregate V20 replay inferred here
Old HHH inclusion–exclusion closure High-quotient HHH Gram	RETRACTED SUPERSEDED AS CONTROLLING	closure proof not source-exhaustive retained formal/provenance infrastructure
Two-high-projector algebra	PASS	exact $AA - BA - AB + BB$ reassembly

Node	Current status	Publication interpretation
Double orthogonality	PASS	exact $W - W' = qt$ and $q \mid 2mW' + s$ geometry
One-dimensional short- t sieve	EXTERNAL PASS	source-specific upper-sieve input
Shiu divisor average	EXTERNAL PASS	audited progression parameter map
Bad-character D^2 ledger	PASS	corrected $C_{\text{var}}(B_0)$ log budget
BALANCED7-OMEGA-FM723-SOURCE-ADAPTER45	SOURCE-OPEN / FIRST RESIDUAL	identify seven physical weights with one Ford-(7.23) packet
Prime-box L^1 normalization	SOURCE-OPEN	print true density/log cost
Shared B_0 cutoff	SOURCE-OPEN	analytic and comparison cutoffs must match
Physical C_{ext}	SOURCE-OPEN	expose all external amplitude-level log powers
Outer-two / inner-five ownership	SOURCE-OPEN	prevent density double-spending
Two-high-projector physical closure	CONDITIONAL	requires all source pins and $C_{\text{var}}(B_0) > 2C_{\text{ext}}$
Inverse-sampled high-conductor LOGVAR	CONDITIONAL / OPEN	not yet closed
Sixth moment fallback	STRONGER SUFFICIENT / OPEN	optional, not controlling
Factorial/physical comparison	SOURCE-OPEN / DOWNSTREAM	becomes controlling only after physical analytic closure
Gate-1B old endpoint-offdiag	SUPERSEDED AS CURRENT FRONTIER	historical endpoint reduction
Gate-1B repaired top-band	RESEARCH REPAIR CANDIDATE	pending source-exact next audit; 1/104 only capacity
Ford generated-(7.23) census / SOURCE-ADAPTER45 / C_{FM}	OPEN	required for proof-specific Type-II replacement
Balanced R_9 binomial 70	FINITE PASS	physical $G(d; n)$ specialization source-open
Remaining fixed- g_* census	OPEN	Balanced7 is only one generated family
FCL	OPEN	fixed-certificate leakage not closed
Gate 2	CONDITIONAL / NOT ACTIVATED	consumes complete provider package, N_2 , G/H , ε_*
Eventual WindowPairSupply	OPEN	needed for global large- M closure
Erdős Problem #287	OPEN	no unconditional solution claim

29 Superseded, false and provenance-only routes

The following material is preserved for auditability but is not allowed to displace the current source-first frontier:

- V15 squarefree polarization plus the $X^{6/7+o(1)}$ repeated-prime router: valid historical representation, superseded by factorial Euler encoding;
- complete multiplicativity as a required repair: retired/not needed;
- direct Pascadi (3/5, 1/7) route: audited nonclosing under the written parameter architecture;
- 3221 DI/Kuznetsov literal-splice lane: superseded/source-mismatched after physical reconstruction;
- direct Kloosterman-fraction black-box lane: audited nonclosing/noncontrolling;
- the post-Poisson demand for an $H = X^{11/35}$ spectral gain: superseded because pre-Poisson divisor density recovers the missing density arithmetically;
- V20 HHH Gram representation: retained as formal infrastructure, but the old HHH inclusion-exclusion *closure proof* is retracted and the HHH cell is superseded as the controlling frontier;

- direct singleton = Gate-1B common-conductor covariance: false as a literal source dictionary;
- direct canonical Gate-1A determinant-one adapter: literal determinant mismatch;
- equal-seven Pascadi Proposition 6.3 three-block dictionary: false at the literal geometry;
- ordinary separated multiplicative large sieve / high moments: nonclosing;
- modulus micro-switch as exhaustive closure: nonexhaustive;
- Gate-1B broad arbitrary-shift beta-weighted operator: sufficient but overstrong as a reduction;
- one beta-side TT^* : exact representation loop;
- uniform $K^{-\delta}$ high- k shortcut: retired.

30 Formal theorem and provenance map

Public Lean bank

- `RequestProject/Erdos287/ProblemStatement.lean`: exact problem predicate and ordered public-statement compiler.
- Finite-range files: adjacent-hole obstruction, prime-power window blocker, and `no_Erdos287Counterexample_of_max_le_4e9`.
- `RequestProject/Erdos287/ClosureInputs.lean`: `WindowPairSupply` and the end-to-end conditional closure compiler.
- V14 source files/status: valid Vaughan structural audit, retained as provenance rather than the controlling source-minimal route.

Lean-proved in the latest supplied V15 replay

The supplied V15 build report records kernel-checked proofs of:

- $\Lambda = \mu * \log$ and its affine specialization;
- exhaustive small- q /small- r /hard recombination;
- the determinant-one line, coprimality, iff parametrization, and uniqueness;
- the finite exponent ledger;
- the balanced-seven binomial coefficient -20 ;
- the repeated-prime finite router;
- the squarefree labelled polarization and its 7^{-7} normalization.

The same status source leaves the analytic endpoint and comparison structures uninhabited.

Public synchronized replay layers and current research frontier

The public aggregate build imports the V15–V19 modules. Their reports record successful builds of 8118, 8123, 8131, 8135 and 8140 jobs, respectively. V20’s HHH source object is public but is treated here as a source/open-interface file rather than an aggregate replay claim.

The current two-high-projector algebra, double-orthogonality reduction, bad-character D^2 ledger, one-dimensional sieve and Shiu average are research/external analytic audits. They do not become a physical theorem until the Ford-(7.23) source adapter, prime-box normalization, shared cutoff and external logarithmic cost are source-pinned.

Gate and Ford–Maynard source provenance

The Gate architecture in Section 4 is a compressed restoration of the earlier standalone verification dossier, not a reconstruction from theorem names alone. The latest supplied formal bank further records: the Gate-1A abstract synchronized two-view fibre challenge $N_\nu = X^{o(1)}$ together with unit-twist portability; the Gate-1B one-completion coefficient $\beta_{D,P} = \mu_D \Lambda_P$, fixed-unit reciprocal-source invariance, the exact $1/104$ dyadic margin, and the open source-to-convolution-BV interface; and the balanced- R_9 finite binomial certificate. The physical Ford $G(d; n)$ specialization, the generated-(7.23) census, the replacement certificate C_{FM} , the Gate-1A necessity decision and all Gate-2 analytic inputs remain open.

Historical technical provenance

The following remain useful diagnostics but are not part of the controlling public proof architecture:

- the squarefree polarization plus the $X^{6/7+o(1)}$ repeated-prime router;
- the V14 Vaughan/prime-modulus Möbius two-outer route;
- failed direct Gate-1A and native-QK dictionaries;
- separate multiplicative-large-sieve and equal-seven three-block death tests.

They belong in the technical verification dossier rather than in the main narrative.

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